

MHF4U - Advanced Functions

Assignment 1 - Function Transformations

Student name 	Date
Total marks 40	Suggested due date September 18, 2026 at 11:59 p.m. ET

- connect equations, graphs, domains, and ranges for transformed functions
- communicate transformations using correct parameters and notation
- justify inverse-function and restriction decisions

Submission instructions

Show all reasoning. Label graphs with scales and key features. You may use graphing technology to verify work, but your submitted solution must include original algebraic reasoning.

Submit one readable PDF through the matching Moodle assignment. Include your name, page numbers, and complete reasoning. Do not submit shared links or temporary draft-file URLs.

1. For $g(x) = -2(x - 3)^2 + 5$, describe every transformation from $y = x^2$. State the vertex, axis of symmetry, domain, and range. 17

2. Sketch $h(x) = 3|x + 2| - 4$ using a table of at least five points. State both intercepts and the range. 17

3. The graph of $y = 1/x$ is transformed to $r(x) = -4/[2(x - 1)] + 3$. Identify a, k, d, and c, then state both asymptotes. /7

4. Find the inverse of $f(x) = (3x - 7)/5$. Verify the result by composition and state the domain and range of each function. /7

5. Create an original transformed function whose graph has a maximum at (2, 6), passes through (4, -2), and opens downward. Determine an equation and justify it. /6

6. Reflection: explain one common transformation error and show how a point-mapping check prevents it. /6

Completion check

Before submitting: confirm every restriction is stated, graphs are labelled, units are included, and the total PDF is readable at 100% zoom.